

Lab 16: Understanding & Implementing Window to Viewport Transformation

Assignment Date: February 2nd, 2026

Deadline: February 10th, 2026

Class: CSIT 3rd Semester, New Summit College

Subject: Computer Graphics

Submission Mode: **Handwritten Document** (Print Output Section only)

Objective: Understanding & Implementing Window to-Viewport transformation

Tasks:

1. Understand the basics of Window to-Viewport transformation
2. Write a Program for Window to Viewport transformation
3. Analyze the output and prepare a conclusion.

Report Format & Structure (Topics to include for submission):

- Index Page
- Theory:
 - a. Window to Viewport transformation
 - b. 2D Viewing Pipeline
- Algorithm of Window to Viewport transformation
- Program for Window to Viewport transformation
- Implementation/Output (You need to label the **Object** and **Image with coordinates** clearly in the output image presented)
- Conclusion

References:

1. <https://drive.google.com/file/d/196P7IHyaJDqC2uy3HWLVLLonl6Ks0fVn/view?usp=sharing>
2. <https://www.geeksforgeeks.org/window-to-viewport-transformation-in-computer-graphics-with-implementation/>